THE ADDRESSABILITY LIMIT: A GEOMETRIC CAPACITY BOUND FOR INFORMATION-GENERATING HIERARCHIES

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Abstract. A process cannot retain distinguishable histories faster than its representation gains room to keep them apart: for retained-history growth β, radial conversion rate c and host packing entropy h_{pack}, $\beta \le c\, h_{\mathrm{pack}}$ in every proper metric space, with no curvature, tree or stationarity hypothesis, and the bound is attained exactly by block codes. What packing entropy does not settle is structure. Over sources with bounded metric balls, a code that preserves genealogy, not only identity, is bounded by packing entropy at *every* scale, so addressing reads growth at the reader's scale and genealogy reads the quasi-isometry-invariant growth; a proper host of entropy $\log 2$ with relational capacity zero exhibits the gap, and bounded geometry closes it. In real hyperbolic space relational capacity attains block capacity, $C^{\mathrm{wt}}_{\mathrm{rel}}(\mathbb{H}^n_\kappa, c; \varepsilon) = c(n-1)\sqrt{\kappa}$. Exact saturation with bounded distortion forces a tree at the reader's scale; a random writer on a finite tape keeps genealogy only inside a window whose two edges are the same inequality at two depths. Efficiency against the bound is a product of five ratios whose content is an experiment. The kernel is machine-checked in Lean 4.

On this manuscript. This paper supersedes a Zenodo record (concept DOI 10.5281/zenodo.19381557) whose public face was an *equality* between curvature and entropy rate; that equality survives only as the named equality case of an inequality (§7), and the content is now the separation between two capacities. The companion, *Evolution as Active Geometry* (Fenn & Fenn), instantiates the bound and ladder on genomic data under its own evidentiary standard; it is cited, not fed back into any theorem below. No biological claim appears here.

1. Introduction, and how to read this paper

A process that generates and retains distinguishable histories needs somewhere to put them. If a history is to stay distinguishable from every other retained one, it must occupy a location no other occupies at the working resolution; and a representation offers only finitely many such locations within a given reach. So unbounded memory inside a bounded radius is not merely hard but impossible—and the central fact of this paper is that the wall is a strict ceiling, standing exactly where the counting puts it.

What kind of statement this is. The result is a *limit principle*: an inequality that cannot be beaten, whose equality case is an idealized realization no particular system is assumed to reach. Its proof is a count, in the way the converse half of a coding theorem is a count; the mathematics that is not a count lies in the separation between addressing and genealogy (§3), in the hyperbolic equality (§5), and in what exact equality forces (§7). The honest premises separating the unconditional statements from the conditional ones are stated as they arise. The old "state equation" is the equality case in an isotropic host, and it appears here in that role only.

The layered structure is the skeleton. The reader can hold the whole paper as five altitudes, and no claim below is stated above its altitude:

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Key words and phrases. Metric packing, information capacity, hyperbolic geometry, coding theorems, volume entropy, critical exponent.

Layer	Statement	Status
I (§2–§3)	Bound $\beta \le c\,h_{\mathrm{pack}}$, min form, exact block identity $C_{\mathrm{block}} = c\,h_{\mathrm{pack}}$; trichotomy; ladder, with rung 2 closed under bounded geometry; the separation of addressing from genealogy, per source and at the level of capacities over metric-ULF sources. Curvature-free.	**Proven** (Lean: finite/convergent case; separation at $\varepsilon = c = 1$ and over the metric-ULF family)
Window (§4)	On a finite tape, a distance reader's identity edge is one-sided and its genealogy edges bound the clock in an explicit interval; readable depth is logarithmic in L at one rate.	**Proven** (Lean); the best-reader line is a named hypothesis
Bridge (§5.1)	Four-point condition $\Leftrightarrow$ tree; $n = 2$ embeddability floor. Neither Layer I nor II.	**Proven** (classical)
Host-specific (§5)	Relational capacity $C^{\mathrm{wt}}_{\mathrm{rel}}(\mathbb{H}^n_\kappa) = c(n-1)\sqrt{\kappa}$; zero tax. *Not* Layer-I universal.	**Proven** (uses [2])
Audit (§5.3)	Efficiency is a product of five ratios, each at most one.	**Proven** (Lean); content is an experiment
IIa (§6)	A3 (full $O(d)$ symmetry) $\Rightarrow M = \mathbb{H}^n_\kappa$; the state equation as equality case.	**Proven** (Schur); A3 *asserted*
IIb (§7)	Exact saturation with bounded tax forces a tree at scale ε (Theorem 3.8, §3); variational selection of the boundary; $\eta = \beta/(ch_{\mathrm{pack}})$. Does a process saturate?	Rigidity and selection **proven**; saturation **open**

Relation to the companion. *Evolution as Active Geometry* imports exactly five objects from this paper and re-derives none: the *bound* of §2 in its min form, the block/relational *ladder* of §2.4 and §5, the *audit identity* of §5.3, the *chart* statements of §5.1 and §6, and the random writer's *window* of §4. Nothing in the companion's empirical content is a premise below.

Formal verification. A theorem tagged [LEAN: `name`] is machine-checked in Lean 4 against Mathlib, with every declaration depending only on `propext`, `Classical.choice` and `Quot.sound`; the tag names the certifying declaration in `theory/lean/ActiveGeometry/` of the repository `sentry-bio/addressability-limit`. The module-by-module inventory, the challenge files (the same statements with every proof replaced by `sorry`) and their job files are in the companion mathematical note [1]. The geometry behind Theorem 5.3's lower bound, the representation theory of Proposition 6.2, Gromov's quasi-isometry invariance, and the theorems of Manning, Katok and Besson–Courtois–Gallot are classical and cited; the Lean predicate `IsotropicHyperbolic` names a hypothesis, not a derived conclusion.

Minimal statement.

$$\boxed{\text{retained exponential novelty} \implies \beta \le c\,h_{\mathrm{pack}}} \qquad \boxed{D_\varepsilon(R_\tau) \le \min\big(|S_\tau|, P(A_\tau, \varepsilon)\big)}$$

unconditionally; separately, $\boxed{\text{A3} \implies M = \mathbb{H}^n_\kappa}$ by Schur, with no claim that any process satisfies A3, and $\boxed{\text{exact saturation with bounded tax} \implies \text{tree at scale } \varepsilon}$. The inequality is the enduring limit; the state equation $\bar\kappa^* = (h_{\mathrm{eff}} \ln 2/(n-1))^2$ is its equality case in the isotropic host, asserted of no system here.

1.1. **Relation to existing work.** The pairing of trees with negatively curved hosts has a substantial literature; it is worth stating which questions it answers and which one it leaves.

Embedding results. Sarkar embeds any finite weighted tree in the hyperbolic plane with edge weights realized exactly and non-adjacent distortion below any $1+\varepsilon$ [15]. Bonk–Schramm embed a Gromov hyperbolic space of bounded growth into $\mathbb{H}^n$ with additive distortion after rescaling [12]. Buyalo–Schroeder embed every Gromov hyperbolic group quasi-isometrically into a product of binary trees, with a sharp factor count [13, 14]. These settle *whether* an object embeds, at what distortion, into how many factors. Our question is different in kind: fixing a resolution ε and a radial budget $d(o, f(v)) \le c\,\tau(v) + A_0$ turns embeddability into a *rate* problem—the supremum of exponential growth rates over admissible codes, not the distortion of any one. The two do not determine each other: low distortion for a fixed tree bounds neither, since a single embedding may consume arbitrary radius. We are not aware of a prior statement that this supremum equals ambient volume entropy exactly (Theorem 5.3), nor of the separation of §3 stated as a capacity rather than a dimension. Kolmogorov–Tikhomirov ε-capacity is the block case without a clock; the radial budget and the generative clock are what make this a rate problem rather than a counting problem.

Critical exponents. The lower bound belongs to the Patterson–Sullivan tradition, in which a group's critical exponent equals volume entropy and, for geometrically finite Kleinian groups, the limit-set Hausdorff dimension [7, 8, 6]. We require a sharper object: a free sub-semigroup with ε-separated orbit whose critical exponent approaches the ambient lattice's, which is exactly Skenderi's construction [2]. Tessera [3] and Kerr [4] give quasi-isometrically embedded free sub-semigroups, hence *positivity* of relational capacity, in broad homogeneous classes; equality is known only in the rank-one case treated here.

Terminology. Capacity always means an exponential growth rate of an admissible code, never the capacity dimension of [13]. *Host* means the ambient target. Dictionary for readers from metric geometry: block address capacity is the growth rate of ε-separated packings of balls; relational capacity restricts to codes whose image is a uniformly quasi-isometrically embedded weighted tree; the generative clock τ is the path-length metric of an edge-weighted tree.

2. The limit

2.1. **Definitions.**

Definition 2.1 (Retained-history growth)**.** Let $\mathcal{H}_R$ be the histories retained through generative depth R, with $\mathcal{H}_R \subseteq \mathcal{H}_{R+1}$, and $N(R) := |\mathcal{H}_R|$. The retained-history growth rate is

$$\beta := \limsup_{R\to\infty} \frac{\log N(R)}{R} \qquad \text{(nats per generative step)}.$$

The word *retained* is load-bearing: a process that overwrites its past, so $\mathcal{H}_R$ does not grow, has $\beta = 0$ regardless of instantaneous change. That is the trichotomy's third horn, not an exception to it. Define the effective bit rate $h_{\text{eff}} := \beta/\ln 2$, so $\beta = h_{\text{eff}} \ln 2$ *by definition*—a unit conversion, not a physical claim. A Shannon entropy-rate estimator measures h_{eff} only when a typical-set or coding argument shows it tracks the exponential growth of retained distinguishable histories; alphabet entropy alone does not establish this.

Definition 2.2 (Faithful representation at finite radial rate)**.** For a pointed metric space (M, d, o), a family $f_R : \mathcal{H}_R \to M$ is *ε-faithful at finite radial rate* when (1) f_R is injective; (2) $d(f_R(x), f_R(y)) \ge \varepsilon$ for distinct x, y; (3) $f_R(\mathcal{H}_R) \subseteq B(o, r(R))$; and (4) $c := \limsup_R r(R)/R < \infty$.

The units of c are geometric length per generative step; it converts the process clock to the host's radial ruler and cannot be omitted when curvatures under different radial gauges are compared. The convention $c = 1$ (the *process-time gauge*) is a coordinate choice, not a theorem.

Definition 2.3 (Packing entropy)**.** With $P(B(o,\rho),\varepsilon)$ the supremum cardinality of an ε-separated subset of $B(o,\rho)$,

$$h_{\mathrm{pack}}(M,o;\varepsilon) := \limsup_{\rho\to\infty} \frac{\log P(B(o,\rho),\varepsilon)}{\rho}.$$

This needs only a metric; it applies to trees, graphs, statistical manifolds, learned representations, and Riemannian manifolds alike. For a complete Riemannian manifold with uniform positive-lower and finite-upper volume bounds on balls of the relevant fixed radii, h_{pack} agrees with volume entropy h_{vol} at exponential order; we use them interchangeably only under that stated condition. The three quantities $\beta, c, h_{\mathrm{pack}}$ are logically independent.
The room's five names. The invariant this paper needs is h_{pack} alone, and it carries five classical names, each under its own hypothesis. In a complete Riemannian manifold of bounded geometry it is the *volume entropy* $h_{\mathrm{vol}} = \lim_{\rho\to\infty} \rho^{-1} \log \mathrm{vol}\, B(o,\rho)$. For a closed negatively curved manifold, Manning's theorem identifies the volume entropy of the universal cover with the *topological entropy* of the geodesic flow [9]. For a discrete group of isometries of a proper geodesic space, orbit growth gives the *critical exponent* $\delta(\Gamma)$ of the Poincaré series, equal to h_{vol} for cocompact lattices [7, 8] and, for geometrically finite Kleinian groups, to the *Hausdorff dimension of the limit set* [6]. The *Liouville entropy* of the geodesic flow is the fifth, and it is the one that comes apart: it agrees with the topological entropy only for locally symmetric metrics in the cases where this is known (surfaces, Katok [10]), and lies strictly below otherwise. Theorem 2.4 uses only h_{pack}, which exists in every metric space, so none of the hypotheses behind the other four is a hypothesis of the bound; they are the names under which the same room has been measured in dynamics, geometry and group theory. Read through Manning's name, the bound says that a lineage cannot branch faster than its medium's geodesics diverge.

2.2. **The bound and its converse.**

Theorem 2.4 (Addressability bound)**.** *If a process of rate β admits an ε-faithful representation at finite radial rate $0 < c < \infty$ in a host of finite packing entropy h_{pack}, with $r(R) \to \infty$, then $\beta \le c\, h_{\mathrm{pack}}$.*

Proof. Faithfulness gives $N(R) \le P(B(o, r(R)), \varepsilon)$. For every $\delta > 0$, eventually $r(R) \le (c + \delta)R$; monotonicity gives $P(B(o, r(R)), \varepsilon) \le P(B(o, (c+\delta)R), \varepsilon)$. Finiteness of the limsup gives $\log P(B(o,\rho),\varepsilon) \le (h_{\mathrm{pack}} + \delta)\rho$ at large radii. Hence $\log N(R)/R \le (c+\delta)(h_{\mathrm{pack}} + \delta) + o(1)$; take the upper limit, then $\delta \downarrow 0$. □

The finite-depth count and the convergent-rate case are machine-checked [Lean: `convergent_rate_addressability_limit`]; the limsup statement is the paper theorem. The converse is not merely existence—it is an exact identity in full generality.

Definition 2.5 (Block address capacity)**.** $A_{\mathrm{block}}(\rho,\varepsilon) := \max\{|C| : C \subseteq B(o,\rho),\ d(x,y) \ge \varepsilon\ \forall x \ne y\}$ and $C_{\mathrm{block}}(c,\varepsilon) := \limsup_R R^{-1} \log A_{\mathrm{block}}(cR,\varepsilon)$.

Theorem 2.6 (Exact block-addressability identity)**.** *For every proper metric space (M,d,o), closed balls are compact, so $A_{\mathrm{block}}(\rho,\varepsilon) = P(B(o,\rho),\varepsilon)$ for all ρ,ε, and consequently $C_{\mathrm{block}}(c,\varepsilon) = c\, h_{\mathrm{pack}}(M,o;\varepsilon)$.*

Proof. Properness gives compact balls, hence finite attained packing numbers; $A_{\mathrm{block}} = P$ is the same extremal problem. Sampling the monotone packing-number function at linearly spaced radii $\rho = cR$ preserves its radial limsup. □

Tightness is what separates a capacity from a constraint: nothing exceeds $c\, h_{\mathrm{pack}}$ and nothing prevents reaching it, so h_{pack} *measures* retained diversity rather than merely bounding it. This requires no tree, hyperbolicity, isotropy, or stationarity. It also means that reaching the envelope is

not, by itself, a signature of anything: block codes reach it in every proper space, and the geometry of a host is forced only by the stronger, exact condition of §7. The finite-radius existence half is machine-checked (`exists_optimal_blockCode` with `card_le_packingCount`).

Corollary 2.7 (Polynomial-growth exclusion; the trichotomy)**.** *If $\beta > 0$ and $c < \infty$, then $h_{\mathrm{pack}} > 0$: no host of only polynomial packing growth can faithfully represent retained exponential novelty at finite radial rate.*

This is the coordinate-free content: *remembering while creating requires an exponential-growth host*—on pain of the address radius failing to grow at any finite rate, or of forgetting. Positive packing entropy does *not* by itself imply negative curvature or Gromov hyperbolicity; hyperbolic space becomes canonical only after the symmetry assumption of §6.

2.3. **The bound in min form.** The rate statement hides the quantity a measurement actually sees. At a single depth the bound is a minimum of two counts, and it prices in correlation and mixture without any independence hypothesis.

Theorem 2.8 (Min form)**.** *Let S_τ be the histories at depth τ, $x : S_\tau \to M$ their images, $R_\tau := x(S_\tau)$, and $A_\tau \supseteq R_\tau$ any region of the host. Let $D_\varepsilon(R_\tau)$ be the number of single-linkage ε-clusters of R_τ (classes of the transitive closure of $d \le \varepsilon$), the number of histories a reader of resolution ε distinguishes. Then*

$$D_\varepsilon(R_\tau) \;\le\; \min\big(|S_\tau|,\; P(R_\tau,\varepsilon)\big) \;\le\; \min\big(|S_\tau|,\; P(A_\tau,\varepsilon)\big).$$

If $|S_\tau| \le e^{\beta\tau}$, $R_\tau \subseteq B(o, c\tau)$, and $\log P(B(o,r),\varepsilon) \le h_{\mathrm{pack}}(\varepsilon)\, r + o(r)$, then $\tau^{-1} \log D_\varepsilon(R_\tau) \le \min(\beta,\, c\, h_{\mathrm{pack}}(\varepsilon)) + o(1)$.

Proof. Pick one point per cluster; points in different clusters are at distance $> \varepsilon$, else they would be linked, so the chosen set is ε-separated in R_τ and $D_\varepsilon(R_\tau) \le P(R_\tau,\varepsilon)$. Monotonicity of P in the set gives the second inequality; $D_\varepsilon \le |S_\tau|$ is trivial. The rate form follows by taking logarithms. □

Three remarks. The middle term $P(R_\tau,\varepsilon)$ counts distinguishable *states*, so correlations among histories are priced in; this is why counts of independent events or channels over-predict retained memory and packing does not. The reader enters only through (d,ε). And the filling fraction $\varphi_\varepsilon(\tau) := D_\varepsilon(R_\tau)/\min(|S_\tau|, P(A_\tau,\varepsilon)) \in [0,1]$ is the single-depth form of the efficiency η of §5.3. The finite statement is machine-checked [LEAN: `distinguishable_le_min`].

2.4. **The admissibility ladder.** Theorem 2.6 proves achievability at exactly one level of task. Stronger senses of "represent a hierarchy" form a ladder, each class $\mathcal{A}$ inducing $C_{\mathcal{A}} \le C_{\mathrm{block}}$: **(1) Block**—terminal messages ε-separated; **(2) Persistent**—one nested address family retains prior messages; **(3) Causal**—parent-to-child addresses obey a per-step motion bound; **(4) Relational**—the representation preserves the source's pairwise metric up to stated distortion (a uniform quasi-isometric embedding). Packing alone proves achievability at level 1; Theorem 2.9 below lifts it to level 2, at the price of a resolution shift in general and at no price under bounded geometry; §5 shows why the gap to level 4 is not automatically zero, and closes it for the canonical host.

The machine-checked kernel states the ladder as a *profile*: a source σ (the growing census of histories, with its own clock), a code κ (addresses and a radial budget), and an obligation $\mathcal{A}$ (a predicate on the pair) are kept apart, obligations are ordered by implication and declared source families by inclusion, and the chain

$$\beta \;\le\; C_{\mathfrak{S},\mathcal{A}}(c) \;\le\; C_{\mathfrak{S},\mathrm{block}}(c) \;=\; c\, h_{\mathrm{pack}}$$

is the *Structured Addressability Limit*, whose first inequality is the packing converse, whose second is obligation monotonicity, and whose equality is constructive depthwise recoding. Two finiteness

conditions on a source are in play and must be kept apart. *Clock-ULF* (Lean `Source.ULF`) bounds the census ratio, $|\sigma(R{+}1)| \le K\,|\sigma(R)|$: finite-rate generation. *Metric-ULF* (Lean `Source.MetricULF`) bounds the balls of the source metric, $|\{v \in \sigma(R) : d_S(u,v) \le t\}| \le f(t)$ uniformly in u and R: bounded valence in the tree case, and the standard meaning of a uniformly locally finite metric space. The binary tree satisfies both; a star with 2^i leaves satisfies the first and not the second. The canonical relational source class is the metric-ULF clocked tree, and the distinction is exactly what separates Corollary 3.4 from Question 3.6. The whole content of the theory lies in whether the middle inequality can be strict.

The second rung is settled up to a resolution shift, and settled outright under bounded geometry. Call a source *retentive* if its censuses are nested, $\sigma(R) \subseteq \sigma(R{+}1)$ (Lean `Source.Retentive`), and call a code *persistent* if an address, once assigned, never moves (Lean `Stable 0`; the theorem holds for any motion bound $m \ge 0$). Write $C_{\mathrm{pers}}(\varepsilon)$ for the capacity over retentive sources under this obligation and $h_{\mathrm{pack}}(\varepsilon)$ for the packing rate of (M,o) at resolution ε when it converges.

Theorem 2.9 (The persistent rung)**.** *Let M be proper, $\varepsilon > 0$, $c > 0$, and let the packing rates at resolutions ε and 2ε converge. Then*

$$c\,h_{\mathrm{pack}}(2\varepsilon) \;\le\; C_{\mathrm{pers}}(\varepsilon) \;\le\; C_{\mathrm{block}}(\varepsilon) \;=\; c\,h_{\mathrm{pack}}(\varepsilon).$$

If moreover the balls of radius 2ε have uniformly bounded packing number at resolution ε (bounded geometry at this pair of scales), then $h_{\mathrm{pack}}(2\varepsilon) = h_{\mathrm{pack}}(\varepsilon)$ and $C_{\mathrm{pers}}(\varepsilon) = C_{\mathrm{block}}(\varepsilon) = c\,h_{\mathrm{pack}}$: the block rate is achieved by one nested address family whose addresses never move.

Proof. Grow a nested family $A_0 \subseteq A_1 \subseteq \cdots$ of ε-separated sets, A_R inside the ball of radius cR, by adjoining to A_R a maximal ε-separated subset of the points of the next ball that lie farther than ε from A_R (Mathlib's maximal separated set of that far region; no Zorn argument is needed). Each A_R is separated, nested in A_{R+1}, and ε-covers its ball, so $|A_R| \ge P(B(o,cR),2\varepsilon)$: a 2ε-separated set injects into any ε-cover by nearest points. Thin each A_R to a nested census of size exactly $P(B(o,cR),2\varepsilon)$, which is nondecreasing in R (a finset between a given subset and a given superset exists at every intermediate cardinality). The identity map is then a base-feasible code at radial rate c that never moves an address, and the census grows at exactly $c\,h_{\mathrm{pack}}(2\varepsilon)$. The upper bound is the packing converse; the bounded-geometry clause is Theorem 3.2. □

[Lean: `capacity_stable_eq_packing_of_boundedGeometry`]

So the price of nesting is at most the gap between two scales of packing entropy, and that gap vanishes in every host of bounded geometry (Theorem 3.2); the balloon, which has no bounded geometry at $(1,2)$, is no counterexample either, since the nested family of all leaves of the first R hubs is 1-separated inside radius R and grows at $\log 2$ (a four-line check, not formalized). Where the ladder costs anything, it costs it at the relational rung.

3. The separation: addresses without genealogy

Packing entropy answers the addressing question completely and the structural one not at all. The reason is a matter of scale: block capacity reads packing entropy at the reader's scale, while a code that preserves genealogy over bounded-valence sources is held to packing entropy at *every* scale. We prove that once, and read the separation off it.

Theorem 3.1 (Coarse packing converse)**.** *Let M be proper, σ metric-ULF with ball bound f, and κ a code satisfying Base at resolution ε and the relational obligation at distortion (D,K). Then for every $\varepsilon' > 0$ and every depth R,*

$$|\sigma(R)| \;\le\; f\big(\lceil D(2\varepsilon' + K)\rceil\big) \cdot P\big(\bar{B}(o,r(R)),\varepsilon'\big),$$

and consequently, if σ has growth β and is relationally representable at radial rate c, then $\beta \le c\,h_{\mathrm{pack}}(M, o; \varepsilon')$ for every ε' at which the packing rate converges:

$$C_{\mathrm{rel}}(M; c, \varepsilon) \;\le\; c \inf_{\varepsilon' > 0} h_{\mathrm{pack}}(M, o; \varepsilon') \quad \textit{over metric-ULF sources.}$$

Proof. A maximal ε'-separated subset Y of the budget ball is an ε'-cover of it with $|Y| = P(\bar{B}(o, r(R)), \varepsilon')$. Assign each history to a point of Y within ε' of its address. Two histories assigned to the same point have addresses within $2\varepsilon'$, so by the lower relational bound their source distance is at most $D(2\varepsilon' + K)$, and a cell holds at most $f(\lceil D(2\varepsilon' + K)\rceil)$ histories. The rate statement follows by taking logarithms, since $f(\cdot)$ is a constant for fixed ε' and $r(R)/R \to c$. □

[LEAN: `coarse_packing_converse`]

Theorem 3.2 (Bounded geometry closes the scale gap)**.** *Let M be proper and $0 < \varepsilon \le \varepsilon'$. If every ball of radius ε' packs at most N_0 points at resolution ε, uniformly in the centre, then $P(\bar{B}(o, \rho), \varepsilon') \le P(\bar{B}(o, \rho), \varepsilon) \le N_0\, P(\bar{B}(o, \rho), \varepsilon')$ for every $\rho \ge 0$, and the packing rates at ε and ε' have the same limit.*

[LEAN: `packingRate_scale_invariant`]

The infimum over scales is the coarse growth rate of M, a quasi-isometry invariant. So the gap between addressing and genealogy is a gap between growth at one scale and growth at all scales, and it can open only in hosts without bounded geometry: the balloon's exponential room is dust below scale 2 and a line above it. In a host whose packing entropy is the same at every scale—every proper geodesic space of bounded geometry, and $\mathbb{H}^n_\kappa$ in particular, where it equals the Hausdorff dimension of the boundary—the coarse converse gives nothing beyond Theorem 2.4, and zero relational tax becomes *possible*; whether it is attained is the achievability question Theorem 5.3 answers for $\mathbb{H}^n_\kappa$. So the gap of Theorem 3.1 is a property of hosts without bounded geometry and of nothing else; here is one.

Definition 3.3 (Balloon)**.** Let B have points $\mathsf{hub}(i)$, $i \in \mathbb{N}$, and $\mathsf{leaf}(i, a)$, $a \in \mathbb{Z}/2^i$, with $d(\mathsf{hub}\, i, \mathsf{hub}\, j) = |i - j|$, $d(\mathsf{hub}\, i, \mathsf{leaf}(j, b)) = |i - j| + 1$, and $d(\mathsf{leaf}(i, a), \mathsf{leaf}(j, b))$ equal to 0 if $i = j, a = b$, to 2 if $i = j, a \ne b$, and to $|i - j| + 2$ otherwise.

B is a ray with a star of 2^i leaves at the ith hub. Its closed balls are finite, so it is proper; $h_{\mathrm{pack}}(B, \mathsf{hub}\, 0; \varepsilon) = \log 2$ for $\varepsilon \le 1$, since the 2^m leaves of hub m are 1-separated inside radius $m + 1$ and a ball of radius ρ holds fewer than $2^{\lfloor \rho \rfloor + 2}$ points; and $h_{\mathrm{pack}}(B; \varepsilon') = 0$ for $\varepsilon' \ge 2$, since separation is strict and above scale 2 a ball packs one point per star. [LEAN: `balloon_coarse_entropy_zero`] The hubs have unbounded degree, so Theorem 3.2 does not apply, and the scale gap opens. Let σ_2 be the rooted binary tree with its unit-edge clock: metric-ULF with $f(t) = 2^t$ (generation-R vertices within tree distance t share their first $R - t$ letters), of growth $\log 2$.

Corollary 3.4 (Separation)**.** (i) *At $\varepsilon = c = 1$ the rate $\log 2$ is achievable for σ_2 in B at the block rung and, for every lawful distortion, not at the relational rung.* (ii) *Over the family $\mathfrak{S}$ of metric-ULF sources, for every lawful distortion, resolution and radial rate, no source of positive growth is relationally representable in B: the relational rate set over $\mathfrak{S}$ lies in $\{0\}$. At $\varepsilon = c = 1$ the block rate set contains $\log 2$ and the relational rate set contains 0, so as numbers*

$$C_{\mathrm{rel}}(B) \;=\; 0 \;<\; \log 2 \;=\; C_{\mathrm{block}}(B).$$

Proof. Block. Send $\{0, 1\}^R$ injectively onto the leaves of $\mathsf{hub}(R)$: distinct leaves are at distance $2 \ge \varepsilon$ inside $\bar{B}(\mathsf{hub}\, 0, R + 1)$, so $r(R)/R \to 1$. *Relational.* Theorem 3.1 at any $\varepsilon' \ge 2$ gives $\beta \le c\, h_{\mathrm{pack}}(B; \varepsilon') = 0$ for every metric-ULF source, σ_2 among them. The one-history source is relationally representable at growth 0, and the right-hand equality is the block identity $C_{\mathrm{block}} = c\, h_{\mathrm{pack}}$ with $h_{\mathrm{pack}}(B; 1) = \log 2$. Only the lower side of the relational sandwich is used, here and everywhere in this paper's separation. □

[LEAN: `balloon_capacity_strict`]

Remark 3.5 (What the separation does and does not say)**.** Corollary 3.4 does not lift to the clock-ULF family. B is itself a tree metric—leaves of a common hub are at distance 2, exactly their distance through the hub—and the identity embedding of B's own spanning tree, whose census $\{\mathsf{leaf}(R,\cdot)\}$ is clock-ULF with $K = 2$ but not metric-ULF, is an isometric relational code of growth $\log 2$. So the balloon is not "room without genealogy"; it is room with *its own* genealogy (a star phylogeny after a rapid radiation, which the companion develops), unable to hold any genealogy of bounded valence. Quartet defect δ (§5.1) is likewise not the capacity slack $c\,h_{\mathrm{pack}} - \beta$: a reticulate source can use every address, a perfect tree very few.

Question 3.6 (Capacity-level separation over clock-ULF sources)**.** *Corollary 3.4 answers the capacity-level question affirmatively over metric-ULF sources, with B as the witness. Over the larger clock-ULF family it remains open: is there a proper metric host with $h_{\mathrm{pack}} > 0$ such that* no *clock-ULF source of growth $c\,h_{\mathrm{pack}}$ is relationally representable at rate c? The balloon is not such a host (Remark 3.5). Bartal–Linial–Mendel–Naor* [5] *prove universality in a neighbouring regime (large subsets of any finite metric embed in an ultrametric at bounded distortion); the balloon's stars are ultrametric and are exactly what such a theorem exploits, which is why unbounded valence is the regime in which a universality theorem could still be in force.*

Remark 3.7 (One obstruction, one route; Manning's inequality)**.** Theorems 3.1 and 3.2 leave one necessary condition for zero relational tax over bounded-valence sources: scale-invariant packing entropy, which bounded geometry supplies. They leave achievability as the open half: a quasi-isometric copy of a full-rate bounded-valence tree in the host. Gromov hyperbolicity with bounded growth is one route to it—Bonk–Schramm's rough-isometric embedding into $\mathbb{H}^n_\kappa$ [12] composed with Theorem 5.3—but it is a route, not a requirement: a regular tree times a line has bounded geometry, is not Gromov hyperbolic, and carries the binary tree at full rate in its first factor. Nor do conjugate points obstruct: a closed manifold negatively curved outside a small region has a universal cover quasi-isometric to a negatively curved one, and whatever tree embeds at full rate in the one embeds at full rate in the other. This is where Manning's theorem sits. It is an inequality before it is an equality: for a closed manifold the volume entropy of the universal cover is at most the topological entropy of the geodesic flow [9], with equality in nonpositive curvature (Freire–Mañé: without conjugate points). It compares endpoints with paths for the host's own flow, and in $\mathbb{H}^n_\kappa$ it is the reason an address is a route, so that Theorem 5.3's equality has a dynamical face. It does not enter the capacity classification, and the two examples above are why: the capacity of codes is governed by growth across scales, which quasi-isometries preserve and conjugate points do not see.

Exact saturation is the other rigid statement of coarse geometry, and it uses the same net.

Theorem 3.8 (Rigidity of exact saturation)**.** *Let $X = x(V_{\le\tau})$ be the image of a tree $T_{\le\tau}$ in a host M, and $A \supseteq X$. Suppose (i) X is a maximal ε-separated subset of A, so that $|X| = P(A,\varepsilon)$ is attained by the image; and (ii) the code has bounded tax: $D^{-1}d_T(u,v) - K \le d(x_u,x_v) \le D\,d_T(u,v) + K$. Then A is quasi-isometric to $T_{\le\tau}$: the map $\pi : A \to V_{\le\tau}$ sending a to a vertex with $d(a,x_{\pi a}) \le \varepsilon$ exists by maximality, is onto, and satisfies*

$$D^{-1}d(a,b) - K - 2\varepsilon \;\le\; d_T(\pi a,\pi b) \;\le\; D\,d(a,b) + K + 2\varepsilon.$$

If moreover A is a geodesic space, A is δ-hyperbolic with δ depending only on D, K, ε.

Proof. A maximal ε-separated subset is an ε-net (if some $a \in A$ were farther than ε from every point of X it could be added without breaking separation), which gives π; $\pi(x_v) = v$ gives surjectivity. For $a, b \in A$ the triangle inequality puts $d(x_{\pi a}, x_{\pi b})$ within 2ε of $d(a,b)$; apply (ii). A tree is 0-hyperbolic, and Gromov hyperbolicity of geodesic spaces is invariant under quasi-isometry with constants depending only on the quasi-isometry constants [20]. □

[LEAN: `Rigidity.quasiIsometric_of_net`]

In words: a lineage that fills its room exactly, with bounded tax, forces the room to have the geometry of a tree at the resolution it is read. This is the one sense in which the equality case is rigid in general. The naive converse—"zero tax only in hyperbolic hosts"—is false: every finite tree embeds isometrically in ℓ_1 and in ℓ_∞^n, which are not hyperbolic; those hosts spend dimension instead, and their packing entropy grows with the tree. The converse must hold packing entropy fixed, which is what Theorem 3.8 does. The product $\mathbb{H}^2 \times \mathbb{H}^2$ hosts zero-tax trees by interleaving its factors without saturating the packing of its balls, so it does not attain exact equality and the theorem does not apply to it.

4. The bridge to finite tapes: the window of a random writer

A tape of L letters has diameter L, so its radial rate is $c = 0$ and the asymptotic bound is silent; what applies on a tape is the min form at a single depth (Theorem 2.8). The companion's question is how long a *random* writer on such a tape honours the one inequality this paper is about—histories that are far apart stay far apart in the record—and its answer is a window in the clock rate with two exits. Both exits are elementary once the reader is fixed, and we record them here because they are theorems of this paper's kind rather than the companion's: the model is the four-letter symmetric channel at eigenvalue decay rate ρ, the reader is a four-point (distance) reader, and the statements are inequalities of real analysis, machine-checked (`Window.lean`).

Proposition 4.1 (Identity edge; one-sided)**.** *Let $p(\rho, \ell) = \frac{1}{4} + \frac{3}{4}e^{-\rho\ell}$ be the probability that two letters at path duration ℓ agree. If N sister pairs at duration ℓ are read on $L \geq \log(N/\delta)/(-\log p)$ sites, every pair is distinct except with probability at most δ [LEAN: `identified_of_sites`]; and if a clock ρ_1 achieves this, so does every $\rho_2 \geq \rho_1$ [LEAN: `identified_mono`]. Identity costs $O(\log N)$ sites and never fails from speed.*

Theorem 4.2 (Genealogy edges; the window)**.** *Let a four-point reader on L sites read a split at depth t with internal branch τ, declaring success when the expected per-site contrast times $\sqrt{L}$ clears a constant $\kappa > 0$: $\kappa \leq \sqrt{L} \cdot \frac{3}{4}e^{-2\rho t}(1 - e^{-\rho\tau})$. Then*

$$\frac{-\log\left(1 - \frac{4\kappa}{3\sqrt{L}}\right)}{\tau} \;\leq\; \rho \;\leq\; \frac{\log\left(\frac{3\sqrt{L}}{4\kappa}\right)}{2t}$$

[LEAN: `window`]; equivalently, the readable depth at rate ρ obeys $t \leq \log(3\sqrt{L}/4\kappa)/(2\rho)$ [LEAN: `readable_depth_le`], logarithmic in the record. The lower edge is the internal branch going unwritten [LEAN: `readable_branch_lower`]; the upper edge is saturation of the deep contrast. Both are the lower side of the relational sandwich, at two depths. The Kesten–Stigum line $b\lambda^2 = 1$ [LEAN: `kestenStigum_iff_rate`] is the ceiling for the best *reader of the same tape and is not derived from this inequality; it remains a named hypothesis (`ReconstructionThreshold`).*

With a spectrum of site rates the logarithm becomes a power of L, with an exponent that depends on the reader; that law and its real-data test belong to the companion (its §5.3).

5. Closing the gap: the hyperbolic host

This theorem is host-specific, not Layer-I universal. Hyperbolic space closes the gap the balloon opens, exactly.

5.1. **The four-point bridge.** For four points form $S_1 = d(a,b) + d(c,d)$, $S_2 = d(a,c) + d(b,d)$, $S_3 = d(a,d) + d(b,c)$. A finite metric is an additive weighted-tree metric exactly when the two largest of S_1, S_2, S_3 agree for every quartet [19]; in the geodesic setting, 0-hyperbolicity is exactly the $\mathbb{R}$-tree condition [20]. The defect δ measures departure from this: it decides whether the qualitative object is a tree, and it determines neither a curvature magnitude nor a growth

class, since a metric can be tree-like by the four-point condition and still have polynomial ball growth. One structural floor follows: a genuinely branching tree cannot embed isometrically in a one-dimensional connected manifold, so $n = 2$ is the minimal smooth hyperbolic ambient dimension for branching trees at low distortion, a statement about embeddability and not a fitted or curvature-derived constant. This bridge is neither Layer I nor Layer II; it motivates the host class and supplies no β, c or curvature.

5.2. **The hyperbolic equality.**

Proposition 5.1 (Finite-depth obstruction)**.** *Fix $\varepsilon > 0$ and $0 < c < \varepsilon$. If every rooted tree T must satisfy $f(T_R) \subseteq B(o, cR)$ for every integer depth R, with $f(\mathrm{root}) = o$ and all distinct vertices ε-separated, the only admissible tree is the one-vertex tree.*

Proof. A child $v \in T_1$ needs $d(o, f(v)) \le c < \varepsilon$; separation needs $d(o, f(v)) \ge \varepsilon$; contradiction. □

This is a mismatch between a physical rate, a fixed resolution, and a clock that charges every edge exactly one unit with no startup allowance—a constraint the addressability theorem never imposed. The correct formulation uses an independently calibrated clock.

Definition 5.2 (Weighted relational code)**.** For a rooted, locally finite tree T with positive edge durations $a(e)$, $\tau(v) := \sum_{e \subset [\mathrm{root}, v]} a(e)$, $T_R^\tau := \{v : \tau(v) \le R\}$, a map $f : V(T) \to \mathbb{H}^n_\kappa$ is a weighted relational code at rate c, resolution ε when $f(\mathrm{root}) = o$; distinct vertices are ε-separated; $D^{-1} d_\tau(u, v) - K \le d(f(u), f(v)) \le D d_\tau(u, v) + K$ for some $D \ge 1, K \ge 0$; and $d(o, f(v)) \le c\,\tau(v) + A_0$ for depth-independent A_0. Then $C^{\mathrm{wt}}_{\mathrm{rel}}(\kappa, n, c; \varepsilon) := \sup_{(T,a,f)} \limsup_R R^{-1} \log |T_R^\tau|$.

Theorem 5.3 (Relational capacity of real hyperbolic space)**.** *For every $\kappa > 0$, $n \ge 2$, $c > 0$, $\varepsilon > 0$,*

$$C^{\mathrm{wt}}_{\mathrm{rel}}(\kappa, n, c; \varepsilon) = c(n-1)\sqrt{\kappa} = C_{\mathrm{block}}(\mathbb{H}^n_\kappa, o; c, \varepsilon).$$

Proof. Write $h := (n-1)\sqrt{\kappa}$. Real hyperbolic space $\mathbb{H}^n_\kappa$ has bounded geometry (uniform two-sided ball-volume bounds, Bishop–Gromov), so its fixed-resolution packing entropy equals its volume entropy at every resolution, $h_{\mathrm{pack}}(\mathbb{H}^n_\kappa, o; \varepsilon) = h_{\mathrm{vol}}(\mathbb{H}^n_\kappa) = h$, and Theorem 2.6 gives $C_{\mathrm{block}}(\mathbb{H}^n_\kappa, o; c, \varepsilon) = ch$. It therefore suffices to prove $C^{\mathrm{wt}}_{\mathrm{rel}} = ch$.

Upper bound. Let (T, a, f) be any weighted relational code. For $v \in T_R^\tau$ the radial condition gives $d(o, f(v)) \le c\tau(v) + A_0 \le cR + A_0$, so $f(T_R^\tau) \subseteq B(o, cR + A_0)$. Distinct vertices are ε-separated, so $|T_R^\tau| \le P(B(o, cR + A_0), \varepsilon)$. Since $\log P(B(o, \rho), \varepsilon) = h\rho + o(\rho)$, the additive offset washes out of the exponential rate, and $\limsup_R R^{-1} \log |T_R^\tau| \le ch$.

Lower bound. Fix a with $0 < a < h$; we construct an admissible code of weighted growth rate $\ge ca$.

(1) A lattice, and its critical exponent. Choose a torsion-free cocompact lattice $\Lambda < \mathrm{Isom}(\mathbb{H}^n_\kappa)$ [17]. Cocompactness gives orbit growth $\#\{\gamma : d(o, \gamma o) \le \rho\} \asymp e^{h\rho}$, so the critical exponent of the Poincaré series equals the volume entropy, $\delta(\Lambda) = h$ [7, 8].

(2) An ε-separated finite-index subgroup. The set $F_\varepsilon := \{\gamma \in \Lambda \setminus \{1\} : d(o, \gamma o) < \varepsilon\}$ is finite by proper discontinuity. As a finitely generated linear group, Λ is residually finite [18], so a finite-index normal $\Lambda_\varepsilon \triangleleft \Lambda$ meets F_ε trivially; by invariance the whole orbit $\Lambda_\varepsilon o$ is ε-separated, and $\delta(\Lambda_\varepsilon) = h$.

(3) A free sub-semigroup of nearly full exponent. $\mathbb{H}^n_\kappa$ is Gromov hyperbolic and its Busemann coarse-cocycle is expanding with $\|\gamma\| = d(o, \gamma o)$ (Skenderi [2], Ex. 2.8), so his Theorem 3.1 applies to Λ_ε at any level $a < h$. It produces a finite $S \subset \Lambda_\varepsilon$ generating a *free* sub-semigroup Σ whose word length is comparable to displacement; with freeness this makes the orbit map of the regular $|S|$-ary word tree a quasi-isometric embedding into $\mathbb{H}^n_\kappa$. Proposition 3.2(4), evaluated at the identity, is the partition inequality $\sum_{s \in S} e^{-a\, d(o, so)} \ge 1$. Since $\Sigma \subset \Lambda_\varepsilon$, its orbit is ε-separated, and freeness sends distinct words to distinct orbit points.

(4) The weighted code. Let T be the word tree of Σ and $f(w) := (s_1 \cdots s_k)o$. Give edge s the duration $a_s := d(o, so)/c$. $f(\text{root}) = o$; distinct vertices are ε-separated by step (3); the orbit map is a quasi-isometry for the word metric and hence, as d_τ is bi-Lipschitz to it, for d_τ; and telescoping with isometry-invariance gives $d(o, f(w)) \le \sum_i d(o, s_i o) = c\,\tau(w)$, so $A_0 = 0$.

(5) The growth rate. Because Σ is free, the weighted counting series factors, $Z(q) = \sum_{k\ge 0} g(q)^k$ with $g(q) = \sum_{s\in S} e^{-qa_s}$, converging iff $g(q) < 1$; the abscissa q^* with $g(q^*) = 1$ is the exponential growth rate of $R \mapsto |T_R^\tau|$. At $q = ca$, $g(ca) = \sum_s e^{-a\,d(o,so)} \ge 1$, so $q^* \ge ca$. Hence $C_{\text{rel}}^{\text{wt}} \ge ca$ for every $a < h$, and with the upper bound $C_{\text{rel}}^{\text{wt}} = ch$. □

Remark 5.4 (The single modern input)**.** Steps (1), (2), (5) are classical. The construction rests on exactly one contemporary result—Skenderi's free sub-semigroup of nearly full critical exponent [2]—and the zero-tax corollary inherits that single point of dependency; no other part of this paper does. Positivity of relational capacity in broader homogeneous classes follows from [3, 4]; equality there is a conjecture, not a theorem.

Corollary 5.5 (Zero relational tax)**.** *The relational tax* $\Gamma := C_{\text{block}} - C_{\text{rel}}^{\text{wt}}$ *is zero at exponential order in real hyperbolic space, as a supremum over weighted-clock, quasi-isometrically embedded tree codes. This establishes neither endpoint attainment, equal-edge achievability, nor zero tax for every prescribed genealogy, and must not be transferred to an arbitrary sequence metric or a host outside this construction.*

5.3. **Slack, efficiency, and the audit.** Write $B := c\,h_{\text{pack}}$ and the efficiency $\eta := \beta/B \le 1$. Under the relational class the block slack decomposes:

$$B - \beta = \underbrace{(B - C_{\text{rel}})}_{\text{relational tax } \Gamma\ge 0} + \underbrace{(C_{\text{rel}} - \beta)}_{\text{utilization slack}\ge 0}, \qquad \eta = \underbrace{\frac{C_{\text{rel}}}{B}}_{\text{availability}} \cdot \underbrace{\frac{\beta}{C_{\text{rel}}}}_{\text{utilization}}.$$

In the weighted hyperbolic class $\Gamma = 0$ (Theorem 5.3), so block efficiency is exactly utilization; outside it, an observed $\eta < 1$ may mix tax with under-utilization, and the decomposition is the correct audit.

Utilization refines further. Between the rate a medium could carry and the rate a reader recovers lie three more stages at which history can be lost: what is written, what is kept, and what remains distinct.

Proposition 5.6 (The audit identity)**.** *For positive rates* $C_{\text{block}} \ge C_{\text{rel}}$ *and* $\beta_{\text{written}}, \beta_{\text{kept}}, \beta_{\text{distinct}}, \beta_{\text{read}}$,

$$\eta = \frac{\beta_{\text{read}}}{C_{\text{block}}} = \frac{C_{\text{rel}}}{C_{\text{block}}} \cdot \frac{\beta_{\text{written}}}{C_{\text{rel}}} \cdot \frac{\beta_{\text{kept}}}{\beta_{\text{written}}} \cdot \frac{\beta_{\text{distinct}}}{\beta_{\text{kept}}} \cdot \frac{\beta_{\text{read}}}{\beta_{\text{distinct}}},$$

and if each stage holds no more than the one before (a lossy *chain), every factor lies in* $(0, 1]$ *and* $\eta \le 1$*. At a single depth the last three factors are images of finite sets under maps, and cardinality cannot grow under a map.*

The identity is algebra and asserts nothing about any medium; it is machine-checked as such [Lean: `Chain.efficiency_eq_product`]. Its content is an experiment: estimate each factor from its own mechanism and check that the product matches the directly measured rate; a residual names a further loss. Two cautions govern that experiment and belong here because they follow from the definitions. First, a count of distinguishable states is not an information: a noisy reader can *increase* the number of distinct images it returns while decreasing the information they carry about the source, so the lossy hypothesis holds for counts only when the reader does not fabricate distinctions, and the information-theoretic form of the chain (mutual information with the source at each stage, which the data-processing inequality orders automatically) is the one to prefer when readers are noisy. Second, by §7, an η whose numerator and denominator come from the same representation is a consistency check, not a measurement of saturation.

A converse locates where a near-capacity code puts its mass.

Theorem 5.7 (Radial concentration)**.** *Let a weighted relational code in $\mathbb{H}^n_\kappa$, with $h = (n-1)\sqrt{\kappa}$, radial rate c, and offset A_0, have growth at least $ch - \Delta$ in the finite form $|T^\tau_R| \geq B^{-1}e^{(ch-\Delta)R}$. For $0 < \delta < 1$ put $E_\delta(R) := \{v \in T^\tau_R : \ d(o, f(v)) \leq (1-\delta)\, c\, \tau(v) + A_0\}$. Then, for a constant $C = C(\varepsilon, n, \kappa, A_0, B)$, $|E_\delta(R)|/|T^\tau_R| \leq C\, e^{-(h\delta c - \Delta)R}$ whenever the growth lower bound holds.*

Proof. The histories in $E_\delta(R)$ are ε-separated inside $B(o, (1-\delta)cR + A_0)$, so $|E_\delta(R)| \leq P(B(o, (1-\delta)cR + A_0), \varepsilon) \leq C_1 e^{h(1-\delta)cR}$. Divide by the growth lower bound. □

At exact capacity the fraction below any band vanishes, so "radius is accumulated information" is a consequence of a hyperbolic host *and* near-capacity operation—a statement about codes, conditional on both, and not an unconditional prediction about any physical system.

6. Which host

6.1. **Isotropic realization.** The packing entropy of §2 need not correspond to a curvature. We add a separate canonicality assumption.

Definition 6.1 (Isotropy convention, Assumption A3)**.** The host M is complete, simply connected, unbounded, and homogeneous negatively curved—by Heintze [16] a solvable $N \rtimes_\varphi \mathbb{R}$ with N nilpotent and φ a positive-real-part derivation of $\mathfrak{n} := \mathrm{Lie}(N)$. The generating process carries no directional structure beyond a real inner product on $\mathfrak{n}$: its symmetry acts as the full orthogonal group $O(d)$, $d := \dim \mathfrak{n}$, by Lie-algebra automorphisms.

Proposition 6.2 (Full symmetry forces isotropy)**.** *Under A3, $\mathfrak{n}$ is abelian and φ a positive scalar multiple of the identity; equivalently $M = \mathbb{H}^n_\kappa$, with $h_{\mathrm{vol}} = (n-1)\sqrt{\kappa}$.*

Proof. Two applications of Schur. The bracket $\Lambda^2\mathfrak{n} \to \mathfrak{n}$ is $O(d)$-equivariant. For $d \neq 3$, $\Lambda^2\mathbb{R}^d$ and $\mathbb{R}^d$ differ in dimension, so $\mathrm{Hom}_{O(d)} = 0$ and the bracket vanishes. At $d = 3$, $\Lambda^2\mathbb{R}^3 \cong \mathbb{R}^3$ as $SO(3)$-modules but with opposite parity under $-\mathrm{Id} \in O(3)$, so the bracket again vanishes; $\mathfrak{n}$ is abelian. Then $\varphi \in \mathrm{End}_{O(d)}(\mathbb{R}^d) = \mathbb{R} \cdot \mathrm{Id}$, and positivity forces a positive scalar. Abelian $\mathfrak{n}$ with scalar dilation is the horospherical presentation of $\mathbb{H}^n_\kappa$. □

Remark 6.3 (The diagnostic it licenses)**.** Heintze's classification does not support "generic": the generic member has non-abelian N or a non-scalar contraction, hence pinched anisotropic curvature. Given A3 exactly, hyperbolic space is the *only* realization. A departure from A3 is therefore diagnostic: reducing the symmetry to a proper subgroup names which member of the rank-one class one is in—$U(n) \subset O(2n)$ the complex hyperbolic boundary, $\mathrm{Sp}(n)\cdot\mathrm{Sp}(1) \subset O(4n)$ the quaternionic, $G_2 \subset SO(7)$ the octonionic. Which invariant tensor survives in a real system's boundary data is a sharper, falsifiable question than whether a fitted curvature looks anomalous. This paper states the diagnostic and does not run it; whether any real process satisfies A3 is an honesty item, not a mathematical gap.

Theorem 6.4 (Curvature floor and the state equation)**.** *Under A3, Theorem 2.4 gives $\beta \leq c(n-1)\sqrt{\kappa}$, i.e. the curvature floor $\kappa \geq (\beta/(c(n-1)))^2$. The unit-invariant form uses $\bar{\kappa} := c^2\kappa$:*

$$\bar{\kappa} \geq \left(\tfrac{h_{\mathrm{eff}} \ln 2}{n-1}\right)^2.$$

7. Saturation: the equality case, and what it forces

The state equation is the *equality case* of the bound in the isotropic host—no more, and asserted of no system here. Two kinds of equality must be kept apart. Equality *at exponential order*, $\beta = c\, h_{\mathrm{pack}}$, is reached by block codes in every proper space (Theorem 2.6) and forces nothing about the host. Equality *exactly*, at every depth, with bounded relational distortion, forces the host's coarse geometry (Theorem 3.8, §3).

Proposition 7.1 (Exponential saturation)**.** *If the limits defining β, c, h_{pack} along $r(R)$ exist and $\log P(B(o,r(R)),\varepsilon) - \log N(R) = o(R)$, then $\beta = c\,h_{\mathrm{pack}}$. Combined with A3 this is the boxed equality*

$$\boxed{\bar{\kappa}^* = \left(\tfrac{h_{\mathrm{eff}} \ln 2}{n-1}\right)^2}$$

the "state equation" of the superseded record, retained here only as this conditional remark.

Remark 7.2 (Symmetry, and what it does not follow from)**.** Three further rigidities are classical and are often confused with the one above. Under A3, Proposition 6.2 leaves one host, so the equality value $\bar{\kappa}^*$ is unique. Among metrics of fixed volume on a closed negatively curved manifold, the locally symmetric metric alone minimizes volume entropy (Besson–Courtois–Gallot [11]). And Liouville entropy equals topological entropy only for locally symmetric metrics where this is known (Katok [10]). Each of these says when the room's five names agree with one another; none says that a process which fills its room must sit in a symmetric host, and Theorem 2.6 shows that it need not. Saturation is a property of codes; symmetry is a property of hosts; A3 is the only bridge, and it is an assumption.

Why the boundary is special, without a dynamics. For *any* admissible-representation objective whose geometric cost is strictly increasing in an attainable host-capacity variable, subject to the lossless constraint $c\,h_{\mathrm{pack}} \geq \beta$, the least-capacity minimizer (if attained) lies exactly on $h^*_{\mathrm{pack}} = \beta/c$. Saturation is the natural answer to that one optimization problem, as a reversible cycle is to Carnot's and a capacity-achieving code to Shannon's: the inequality is the limit, equality its ideal realization. Reaching it is a design act. Codes laid down as nets reach it; nothing here says a process that writes at random does.

What is not proved, and what would test saturation. The mismatch $V(\kappa,\kappa^*) = (\sqrt{\kappa} - \sqrt{\kappa^*})^2$ has a unique zero at κ^* and nothing more is claimed of it: no dynamics is supplied, so neither stability nor attraction is asserted. Saturation is tested only by an $\eta = \beta/(c\,h_{\mathrm{pack}})$ whose numerator and denominator have independent provenance; an η produced by fitting one axis to the other, or by back-solving n or κ from an assumed equality, tests nothing.

8. The firewall, and what is open

Theorems 2.4, 2.6, 2.8, 3.1, 3.2, 3.8, 4.2 and 5.3, Corollaries 2.7 and 3.4, and Proposition 4.1 are unconditional. Theorem 6.4 is conditional only on Assumption A3. Nothing above requires that any physical process fills the capacity it proves exists, reports a curvature, back-solves a dimension, or treats an experiment as forcing hyperbolic realization over an alternative exponential host. We do not assert $h_{\mathrm{vol}} = h_{\mathrm{eff}} \ln 2$ as forced by the bound's own axioms; we use no interior-entropy or unimodal-channel hypothesis; and we offer no second derivation from Fisher–Rao considerations.

Three questions. *Mathematical, first:* for which proper metric hosts is packing entropy a complete invariant for relational capacity? $\mathbb{H}^n_\kappa$ is an affirmative answer; Corollary 3.4 gives a negative answer at the level of capacities over metric-ULF sources, and Question 3.6 asks whether one exists over clock-ULF sources, where the balloon's own star genealogy embeds. The first tractable step is the rank-one symmetric family, where positivity is known [3, 4] and equality is not. (Rung 2 of the ladder is no longer open in the bounded-geometry class, Theorem 2.9; elsewhere it is pinned between $c\,h_{\mathrm{pack}}(2\varepsilon)$ and $c\,h_{\mathrm{pack}}(\varepsilon)$, the same scale gap as Theorem 3.1.) *Mathematical, second:* Theorem 4.2 covers both edges of the random writer's window for a *distance* reader, at the level of the model's expected contrast. The corresponding statement for the *best* reader—the erasure edge then being the Kesten–Stigum second-moment line and the collision edge the recurrence threshold of branching diffusion—remains the conjecture the companion motivates; the two readers' edges are different quantities, and this paper does not attempt the second. The

collision edge already has an exact form in one host: branching Brownian motion in $\mathbb{H}^n_\kappa$ reaches the whole boundary if and only if its branching rate exceeds $(n-1)^2/4$, the square of half the volume entropy and the bottom of the L^2 spectrum [22, 23], below which the surviving lineages reach a boundary set of dimension fixed by the branching rate through Sullivan's quadratic. That is a half-entropy threshold of the same shape as the Kesten–Stigum line; the exact identification between the channel's eigenvalue and the heat kernel's decay is the derivation we would attempt first.

Empirical, and untouched here: the bound proves a ceiling exists and the identity proves it attainable; neither shows any physical process approaches it. Whether retained diversity runs near $c\,h_{\mathrm{pack}}$ or orders of magnitude below is a measurement, requiring β and h_{pack} of independent provenance. We regard this as the more consequential question, the one the mathematics is meant to make askable, and one this paper does not answer.

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